
🧪 AWS VPC Endpoint Lab

🎯 Objective

Enable **private access** to AWS services from EC2 instances in a **private subnet** using:

- **Gateway Endpoint** for Amazon S3
- **Interface Endpoint** for AWS Secrets Manager

🧱 Architecture Overview

```
              ┌──────────────────┐
              │   VPC       │
              │ 10.0.0.0/16│
              └────────┬─────────┘
                │
         ┌─────────────────────┐ │ ┌──────────────────────────┐
         │ Public Subnet │───┘   │  Private Subnet    │
         │ 10.0.1.0/24   │     │  10.0.2.0/24     │
         └─────────────────────┘   └────────────┬─────────────┘
            │                │
      [ EC2-A: Bastion ]       [ EC2-B: No public IP ]
                       │
                 ┌────────────▼────────────┐
                 │ Gateway Endpoint    │
                 │    (S3)        │
                 └─────────────────────────┘
                 ┌────────────▼────────────┐
                 │ Interface Endpoint │
                 │ (Secrets Manager) │
                 └─────────────────────────┘
```

🛠️ Step-by-Step Instructions

✅ Part 1: VPC & Subnets Setup

1. **Create a VPC**
    - Name: `VPC-Endpoint-Lab`
    - CIDR block: `10.0.0.0/16`
2. **Create Subnets**
    - `PublicSubnet`: `10.0.1.0/24`
    - `PrivateSubnet`: `10.0.2.0/24`
3. **Create & Attach Internet Gateway**
    - Name: `IGW-Lab`
    - Attach to your VPC
4. **Configure Route Tables**
    - Public Route Table:
        - Add route: `0.0.0.0/0 → Internet Gateway`
        - Associate with `PublicSubnet`
    - Private Route Table:
        - No internet route yet
        - Associate with `PrivateSubnet`

✅ Part 2: EC2 Launch & Test

1. Launch **EC2-A** (Public Subnet)
    - Amazon Linux 2
    - Enable public IP
    - Security Group: allow SSH (port 22)
2. Launch **EC2-B** (Private Subnet)
    - Amazon Linux 2
    - No public IP
    - Same key pair
    - Security Group: allow SSH from EC2-A and allow HTTPS (port 443)
3. **SSH Access**
    - SSH into EC2-A
    - From EC2-A, SSH into EC2-B

✅ Part 3: Gateway Endpoint (for S3)

Step 1: Create S3 Bucket
aws s3 mb s3://vpc-endpoint-lab-<your-name>

Step 2: Try S3 Access from EC2-B (Fails)
aws s3 ls

🔴 Expect error due to no internet or NAT.

Step 3: Create Gateway Endpoint for S3

1. Go to: **VPC** → **Endpoints** → **Create Endpoint**
2. Name: `S3GatewayEndpoint`

3. Service: `com.amazonaws.<region>.s3`
4. Type: **Gateway**
5. VPC: `VPC-Endpoint-Lab`
6. Attach to **Private Route Table**

✅ Adds automatic route via prefix list for S3.

Step 4: Try S3 Access Again from EC2-B

echo "test" > test.txt
aws s3 cp test.txt s3://vpc-endpoint-lab-<your-name>/

✅ Should now succeed via **private connection**.

✅ Part 4: Interface Endpoint (for Secrets Manager)

Step 1: Create a Secret

1. Go to **Secrets Manager** → **Store new secret**
   - Secret name: `MyTestSecret`
   - Key: `username`, Value: `admin`

Step 2: Create Interface Endpoint

1. Go to **VPC** → **Endpoints** → **Create Endpoint**
2. Name: `SecretsManagerEndpoint`
3. Service: `com.amazonaws.<region>.secretsmanager`
4. Type: **Interface**
5. Subnet: `PrivateSubnet`

6. Enable **Private DNS**
7. Attach security group (allow HTTPS from EC2-B)

Step 3: Test Secrets Manager from EC2-B
aws secretsmanager get-secret-value --secret-id MyTestSecret

✅ Should succeed via VPC interface endpoint.

🧼 Cleanup (Optional)

- Terminate EC2 instances
- Delete endpoints
- Delete S3 bucket
- Delete Secrets Manager secret
- Delete VPC

✅ Summary Table

Component	VPC Endpoint Type	Traffic	Notes
Amazon S3	Gateway	Routed	Uses route table + prefix
AWS Secrets Manager	Interface	ENI	Uses private IP & DNS
